

PRACTICAL 1: NETWORK TOOLS AND SAFETY

Activities

1. Identify Networking Tools

- Crimping tool
- Punch down tool
- Cable tester
- Screwdrivers

2. Explain the Use of Each Tool

- **Crimping Tool:** The crimping tool is used to crimp CAT cables with RJ-45 connectors to join them to each other.
- **Punch Down Tool:** The punch down tool is used to place/push individual wires of a cable into slots on patch panels.
- **Cable Tester:** Is used to check the ordering of the wires on a network cable and it indicates if there is a fault in the cable by sending a signal through each wire in the cable.
- **Screwdrivers:** A screwdriver is a basic tool but is still used in networking. They are used to tighten or loosen any screws on network devices or they are used to mount network devices to the wall like routers.

3. Demonstrate Safe Handling

- **Crimping Tool:** Make sure you have good grip on the crimping tool.
- **Crimping Tool Safety:** Make sure you keep your fingers away from the blade at all times.
- **Punch Down Tool:** Do not press too hard on the wires to avoid damaging the wires.
- **Punch Down Tool Safety:** Use safety glasses when operating punch down tools.
- **Cable Tester:** Do not connect it to live wires.
- **Cable Tester Safety:** Handle the cables with care to avoid damaging them.
- **Screwdrivers:** Use the correct size screwdriver to prevent it from slipping.
- **Screwdriver Safety:** Don't use broken screwdrivers.

4. Complete a Tool Inventory Sheet

Toolbox		Working
Lock		Good
Ball of string		Weak
Crossover Category 5e cable, 6 feet		Weak
Flashlight		Good
Fish tape		Good
Measuring tape		Good
Needle nose pliers		Weak

Network cable tester		Working
Nut driver set		Good
Punch down tool	✓	Good
RJ45/RJ11 cable crimper		Good
Safety goggles	X	N/A
Scissors	✓	Working
Screwdrivers	✓	Good
Soldering iron and accessories	✓	Good
Straight thru Category 5e cable, 6 feet	✓	Good
Stud finder	✓	Good
Two-way radio set	✓	Working
Utility knife	✓	Working
Wire strippers	✓	Good

Expendable Materials / Materials Inventory

Assorted wood/sheet screws in a small container	100
Category 5e RJ-45 connectors	40
Nylon wire ties	2 rolls
Electrical tape and duct tape	100
Reg/11 labels	20
Trash bag	10
Wire numbers set	1 set

5. Apply Health and Safety Procedures

- Read the appropriate manufacturer's instructions.
- Identify potential hazards:
 - Loose cables
 - Sharp tools
 - Nearby electrical equipment
- Wear the appropriate personal protective equipment (PPE):
 - Safety goggles
 - Protective gloves
 - Safety boots
 - Dust mask
- Keep your work area clean and organized.
- Don't use faulty equipment.

Deliverable & Assessment

Deliverable: Completed Tool Identification Worksheet